

Household Battery Trading Optimisation in the Australian National Electricity Market

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Chapter 1

Introduction

1.1 Background

The Australian National Electricity Market (NEM) supplies electricity to customers across Australia in NSW, Victoria, Tasmania, the ACT and South Australia. In the NEM, wholesale electricity prices are determined in 5 minute intervals. Households have traditionally navigated the NEM through electricity retailers that charge a fixed or time of use based price per kWh. This project will explore options to optimise returns for households with solar and battery systems participating in the NEM at wholesale prices.

1.2 National Electricity Market

The wholesale of electricity in the National Electricity Market consists of a spot market, where prices are determined in real time. Each region (state/territory) is an individual market with a spot price. The market consists of a 5-minute dispatch period. Energy generators bid a volume of capacity and a price at which to serve the market for the dispatch period.

Using the bids from generators, the Australian Energy Market Operator (AEMO) will use the National Electricity Market Dispatch Engine to order bids from least to most expensive and determine which generators will be dispatched. AEMO's objective is to dispatch the lowest cost generation to meet expected demand [5]. The highest price paid for wholesale electricity is paid to all generators instructed to dispatch by AEMO.

The market mechanisms consist of a Market Price Cap (MPC) of \$20,300/MWh and a cumulative price threshold (CPT) of \$1,823,600, which limits the sum of the spot prices over any seven consecutive days of trading, for 1 July 2025 to 30 June 2026. Both values are indexed to CPI each year by the Australian Energy Market Commission [6]. The simulations

in this report use January 2025, which falls in the 2024–25 financial year, when the MPC was \$17,500/MWh. There are additional markets for ancillary services [7] that will not be addressed in this report but have the potential to be revenue streams for households participating in Virtual Power Plants and are regularly referenced in related literature.

1.3 Motivation

Australia has one of the highest rates of solar uptake in the world, with over 4.3 million households having rooftop solar [8]. The rapid price reduction of home battery systems and the introduction of the Federal Government’s Cheaper Home Battery scheme, have also driven a dramatic increase in battery installations. In the second half of 2025, 183,245 home batteries were installed in Australia [8]. The increase in consumer energy resources (CER) across the NEM is creating both opportunities for customers and challenges for utilities and system operators.

Consumers are increasingly playing prosumer roles in the NEM: consuming, producing, and selling electricity. This has opened opportunities for households to play an active role in the market, engaging with the NEM to minimise electricity costs. However, it has also placed stress on the distribution networks, with consumer energy resources being largely uncontrolled and unresponsive to market signals. The continued uptake of electric vehicles (EVs) also presents an opportunity for consumers to play a more active role in their consumption of electricity. Through exposure to market pricing signals, consumers can engage in energy demand response (DR) by flexing charge rates of EVs. Demand response offers households the ability to self consume more solar, reducing grid import costs, and reduce strain on distribution networks.

This opportunity for market exposure has been constrained by the complexity of the NEM. Mayer et al. [9], in a study of 28 power markets across Asia, Australia, Europe and North America categorised the NEM as ‘by far the most volatile markets considered in this study’. The NEM’s volatility offers an opportunity for more efficient utilisation of CER but also exposes consumers to risk from price peaks.

Chapter 2

System Model

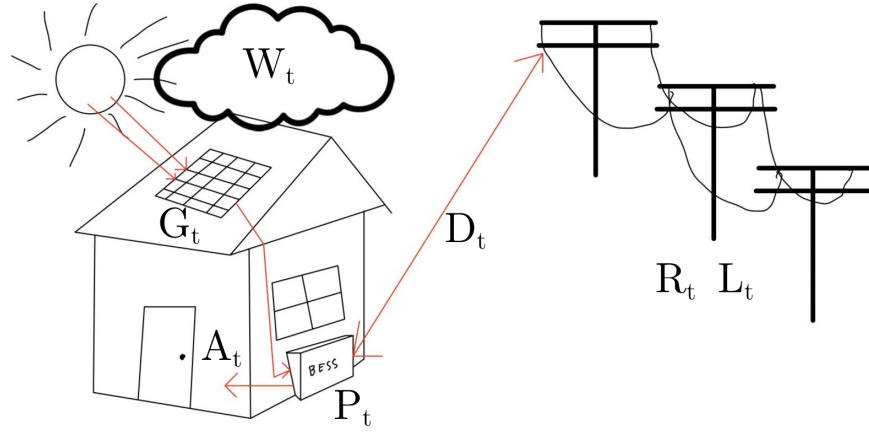


Figure 2.1: A household with solar and batteries connected to the grid

Figure 2.1 shows the basic system model that will be considered.

2.1 Nomenclature

W_t Weather variables t

G_t Rooftop solar generation t (kW)

A_t Household electricity consumption t (kW)

D_t^i Grid power imported (kW)

D_t^e Grid power exported (kW)

P_t^c Battery charging power t (kW);

- P_t^d Battery discharging power t (kW);
- E_t State of charge of battery t (kWh)
- R_t NEM Regional Reference Price t (\$/MWh)
- L_t Total demand in NEM region t (MW)
- C_t^g Energy cost t (\$)
- C_t^b Battery degradation cost t (\$)
- Δt NEM dispatch period, 5 minutes
- N Network tariff (\$/kWh)
- P_{max}^c, P_{max}^d The maximum charging and discharging rates of the battery inverter (kW)
- E^{rate} The rated energy capacity of the battery (kWh)
- E_{max} The maximum permitted state of charge (kWh)
- E_{min} The minimum permitted state of charge (kWh)
- E_0 The state of charge at the start of the horizon (kWh)
- η^c Battery charging efficiency (%)
- η^d Battery discharging efficiency (%)
- α_d Battery degradation coefficient (\$/kWh)
- J Number of cycle depth segments, indexed by j
- δ Cycle depth, as a fraction of rated capacity
- $\Phi(\delta)$ Cycle depth stress function (fraction of cell life lost per cycle of depth δ)
- R^{cell} Battery replacement cost (\$)
- c_j Marginal aging cost of cycle depth segment j (\$/kWh)
- E_j Energy capacity of cycle depth segment j (kWh)
- $P_{t,j}^c$ Battery charging power allocated to segment j at t (kW)
- $P_{t,j}^d$ Battery discharging power allocated to segment j at t (kW)
- $E_{t,j}$ Energy stored in cycle depth segment j at t (kWh)

2.2 Objective

The objective of the project is to minimise household expenditure on electricity consumption while considering battery degradation as defined in Equation 2.1. This objective will be verified through simulations in Python using historical data.

$$\min \sum_{t=0}^T C_t^g + C_t^b \quad (2.1)$$

2.2.1 Cost of Energy

The cost of energy is determined by the wholesale price, quantity traded, and network tariff. Network tariffs are only charged on electricity imported from the grid. The cost of energy can be modelled through Equation 2.2.

$$C_t^g = D_t^i(N + R_t)\Delta t - D_t^e R_t \Delta t \quad (2.2)$$

2.2.2 Battery Degradation

Modelling battery degradation accurately is a highly non-linear process [10]. The dominant operational stress factor is cycle depth: a cell cycled shallowly sustains far more cycles, and delivers far more lifetime energy throughput, than the same cell cycled deeply [10]. A degradation cost that is linear in energy throughput cannot represent this, because it charges the same amount for many shallow cycles as for one deep cycle of equal throughput.

Let $\Phi(\delta)$ denote the cycle depth stress function, giving the fraction of cell life consumed by one full cycle of depth $\delta \in [0, 1]$. Following Xu et al. [11], Φ is approximated by a piecewise linear function of J segments that evenly divide the cycle depth range. Prorating the battery replacement cost R^{cell} over the life consumed by each segment gives the marginal aging cost of segment j :

$$c_j = \frac{R^{\text{cell}}}{\eta^d E^{\text{rate}}} J \left[\Phi\left(\frac{j}{J}\right) - \Phi\left(\frac{j-1}{J}\right) \right] \quad (2.3)$$

Because Φ is convex, $c_1 \leq c_2 \leq \dots \leq c_J$, so a deeper segment is never cheaper than a shallower one. This convexity is what allows the model to be solved as a linear program: the optimiser always discharges the cheapest available segment first, which reproduces the cycle depths that the rainflow counting algorithm would identify without requiring that algorithm inside the optimisation [11]. Following [11], aging is attributed to the discharging half cycle only, on the basis that the energy charged and discharged over a day are almost identical; charging is therefore not separately penalised.

The degradation cost at t is the cost of the discharge power drawn from each segment:

$$C_t^b = \sum_{j=1}^J c_j P_{t,j}^d \Delta t \quad (2.4)$$

Setting $J = 1$ recovers a single constant marginal cost $c_1 = \alpha_d$, which is the linear degradation model. The number of segments J is therefore a fidelity parameter rather than a change of model, and the effect of degradation model fidelity on dispatch can be evaluated by varying J , as proposed in Section 9.2.

2.2.3 Objective Function

The objective is to determine values for $P_{t,j}^c$ and $P_{t,j}^d$ as stated in Equation 2.5. The constraints are discussed in the following section.

$$\min_{P_{t,j}^c, P_{t,j}^d} \sum_{t=0}^T \left[D_t^i (N + R_t) \Delta t - D_t^e R_t \Delta t + \sum_{j=1}^J c_j P_{t,j}^d \Delta t \right] \quad (2.5)$$

$$\text{s.t.} \quad (2.6)$$

$$(2.7)$$

$$(2.8)$$

2.3 Constraints

2.3.1 Battery Storage Model

The battery model introduces constraints to the system as outlined in Equation 2.6. The battery state of charge (SoC) is described in Equation 2.6a. Battery efficiency is a non-linear characteristic of the battery, conditions and SoC. To reduce complexity, this project will initially model a fixed efficiency however there is scope to more accurately optimise the operation.

u_t^c and u_t^d are binary variables. If the battery is charging $u_t^c = 1$ otherwise $u_t^c = 0$, u_t^d follows the same logic for battery discharge. These binary variables are introduced to prevent the model from charging and discharging the battery simultaneously as used in [12].

Constraints 2.6d and 2.6e reflect the battery inverter limits for charging and discharging the battery, which need not be equal (Table 3.1). 2.6i keeps the battery SoC between its minimum and maximum permitted levels, which lie inside the rated capacity E^{rate} .

To represent cycle depth, the battery energy is disaggregated across the J cycle depth segments introduced in Section 2.2.2. Each segment j holds its own energy level $E_{t,j}$,

bounded by the segment capacity $E_j = E^{\text{rate}}/J$, and the total charge and discharge power is the sum of the per-segment components, (2.6b) and (2.6c). Tracking energy per segment, (2.6a), allows the current cycle depth to be identified: because the marginal cost curve is convex, the battery discharges from the shallowest available segment towards the deeper, more expensive ones [11]. The aggregate SoC is $E_t = \sum_j E_{t,j}$.

Constraint 2.6k requires the battery to finish the horizon with at least the energy it started with. Without it the model would discharge the battery in the final intervals regardless of subsequent prices, an end-of-horizon artefact that would overstate revenue.

$$E_{t,j} = E_{t-1,j} + P_{t,j}^c \eta^c \Delta t - \frac{P_{t,j}^d}{\eta^d} \Delta t \quad (2.6a)$$

$$P_t^c = \sum_{j=1}^J P_{t,j}^c \quad (2.6b)$$

$$P_t^d = \sum_{j=1}^J P_{t,j}^d \quad (2.6c)$$

$$0 \leq P_t^c \leq u_t^c P_{max}^c \quad (2.6d)$$

$$0 \leq P_t^d \leq u_t^d P_{max}^d \quad (2.6e)$$

$$0 \leq u_t^c + u_t^d \leq 1 \quad (2.6f)$$

$$u_t^c, u_t^d \in \{0, 1\} \quad (2.6g)$$

$$0 \leq E_{t,j} \leq E_j \quad (2.6h)$$

$$E_{min} \leq \sum_{j=1}^J E_{t,j} \leq E_{max} \quad (2.6i)$$

$$\sum_{j=1}^J E_{0,j} = E_0 \quad (2.6j)$$

$$\sum_{j=1}^J E_{T,j} \geq E_0 \quad (2.6k)$$

2.3.2 Unidirectional Power Flow

Solar generation and household consumption is restricted to unidirectional power transfer.

$$A_t \geq 0 \quad (2.7a)$$

$$G_t \geq 0 \quad (2.7b)$$

2.3.3 Meet Demand

The constraint defined in Equation 2.8 ensures that household demand is always met.

$$A_t = P_t^d - P_t^c + G_t + D_t^i - D_t^e \quad (2.8)$$

2.3.4 Observable State Space

In this project, the household uses a smart electricity meter. Smart meters record the net household electricity consumption at 5-minute resolution. The values of solar generation, G , and household load, A , are not recorded independently. Instead, the net household electricity, H , is used for the optimisation.

$$H = G - A \quad (2.9)$$

A simplifying assumption made in this study is that the net household electricity consumption is known at beginning of each time slot n . This assumption is made to reflect the inverter's ability to respond to real-time changes in household demand and generation. An analysis of the effects of removing this assumption is detailed in Section 8.6.

The electricity spot price is also revealed to the market by AEMO at the beginning of each time slot n . The regional system demand L published by AEMO is likewise available for all time-slots before slot n . The regional system demand is the measured generation in the region adjusted for interconnector flows, with a forecast adjustment for the lag between metering and dispatch [13].

The charge and discharge decisions at slot n are functions of the system observations available at time-slot n , (2.10a), and the history of previous charge/discharge decisions, (2.10b).

$$\mathbf{V}_n = \begin{bmatrix} H_1 & H_2 & \cdots & H_n \\ L_1 & L_2 & \cdots & L_n \\ R_1 & R_2 & \cdots & R_n \end{bmatrix} \quad (2.10a)$$

$$\mathbf{K}_n = \begin{bmatrix} P_1^c & P_2^c & \cdots & P_{n-1}^c \\ P_1^d & P_2^d & \cdots & P_{n-1}^d \end{bmatrix} \quad (2.10b)$$

The charge and discharge power at each time slot n are determined by a policy that

maps the available system observations $\mathbf{V}_n$ and prior decisions $\mathbf{K}_n$ to a dispatch decision:

$$P_n^d = f_{d,n}(\mathbf{V}_n, \mathbf{K}_n) \quad (2.11a)$$

$$P_n^c = f_{c,n}(\mathbf{V}_n, \mathbf{K}_n) \quad (2.11b)$$

2.4 Solution Formulation

In this work the policy has two stages. A forecaster maps $\mathbf{V}_n$ to predicted prices $\hat{R}_{n+1}, \dots, \hat{R}_{n+k-1}$ and predicted net household power $\hat{H}_n, \dots, \hat{H}_{n+k-1}$ over a horizon of k slots. The system will then take the forecast and optimise the selection of P_t^c, P_t^d .

The forecaster will use statistical methods for time series forecasting and recurrent neural networks such as Long Short Term Memory (LSTM). The optimisation will use linear programming and reinforcement learning.

Chapter 3

Technical Background

3.1 Machine Learning

3.1.1 Artificial Neural Networks

Artificial neural networks (ANN) form the basis for the machine learning models that will be used for predicting states and making trading decisions. An ANN consists of layers of nodes connected to one another. Each node produces an output based on the inputs to the node and the tuning parameters; weights and biases. A neural network consists of:

- Input Layers: Holds the raw input to the system. For example, the RRP in the NEM for the previous 24 hours.
- Hidden Layers: Consist of nodes that transform inputs into new representations using linear transformation.
- Output Layers: Transform the output from the hidden layers to a final state using the nonlinear activation function. For example, the predicted RRP in the NEM.

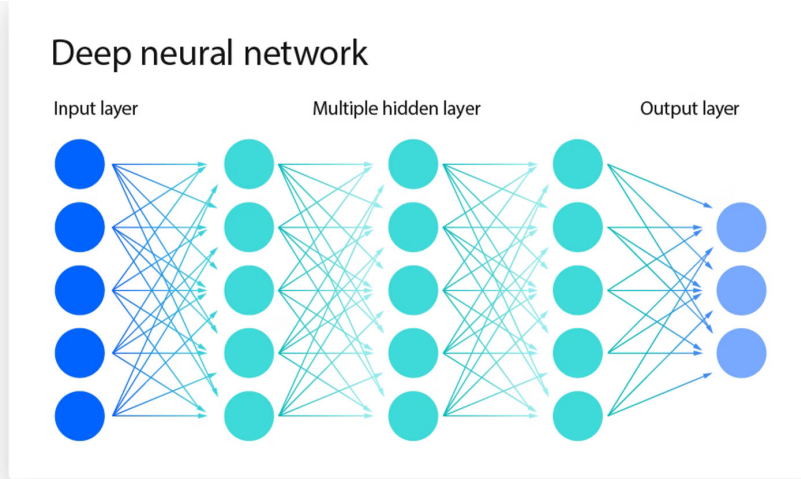


Figure 3.1: A standard feedforward neural network with 3 hidden layers [1].

A single neuron computes a weighted sum, z , and an output, a [1]:

$$z = \sum_{i=1}^n w_i x_i + b \quad (3.1)$$

$$a = \sigma(z) \quad (3.2)$$

Where

- x_i = input feature,
- w_i = weight,
- b = bias,
- σ = activation function (nonlinear transformation),

The weights and biases of the ANN are adjusted through comparing the network's prediction, $\hat{Y}$, to the true label, Y , and measuring the error using a loss function. To minimise this loss function, backpropagation is used to train the neural network. Backpropagation consists of four steps:

1. Forward pass: Inputs flow through the network, computing outputs at each node and producing an output prediction.
2. Error calculation: The error between the expected and actual output is calculated using the loss function.
3. Backward pass (back propagation): The error is propagated back through the network. For each neuron the algorithm calculates the contribution of the weight and bias to the error using the chain rule of calculus.

4. **Weight update:** Using gradient descent methods, the weights and biases are adjusted slightly to reduce the error.

This process is repeated many times over a training set to tune the internal parameters of the network. With appropriate training the network converges to minimise the loss function. An example for this project would be the minimisation of error for predicting NEM RRP.

3.1.2 Long Short Term Memory

Long Short Term Memory (LSTM) networks are a type of recurrent neural network (RNN) designed by Hochreiter and Schmidhuber [14]. RNNs are commonly used ML models for state prediction such as price forecasting. They contain a specific type of neuron called a recurrent neuron. Recurrent neurons have a hidden state that holds information about the previous inputs. Through using Backpropagation Through Time (BPTT), the weights of the network are updated to minimise error across timestamps.

LSTM introduces a memory cell that holds information over extended periods. This acts as long-term memory for the network and assists with pattern recognition. Each memory cell has three gates:

- **Input Gate:** Controls the information added to the memory cell.
- **Forget Gate:** Determines what information is removed from memory cell.
- **Output Gate:** Controls information output from memory cell.

3.1.3 Reinforcement Learning

This project will consider a Reinforcement Learning (RL) approach for controlling household battery operation. RL has shown benefits over MILP when considering the non-linear characteristics of battery degradation and charging efficiencies. The model will take in predicted states, $\hat{G}$, $\hat{A}$, $\hat{P}$, and output an action.

Markov Decision Process

A Markov decision process (MDP) is a mathematical model for sequential decision making when outcomes are uncertain. It is a method to define the problem the RL model will navigate. A MDP is defined by several properties [15]:

- A set of states, S .
- A set of actions, A .

- A start and terminal state(s).
- A transition function, $T(s, a, s')$. Actions in an MDP may be nondeterministic; the transition function represents the probability of an agent taking an action $a \in A$ from a state $s \in S$ and ending up in a state $s' \in S$.
- A reward function, $R(s, a, s')$. This reward may be positive or negative.

Figure 3.2 illustrates a basic MDP with three states. The transition function is represented by the probabilities indicated on each action branch and the agent is rewarded based on the states visited. We define the optimal policy function to minimise the expected cost to the goal as π .

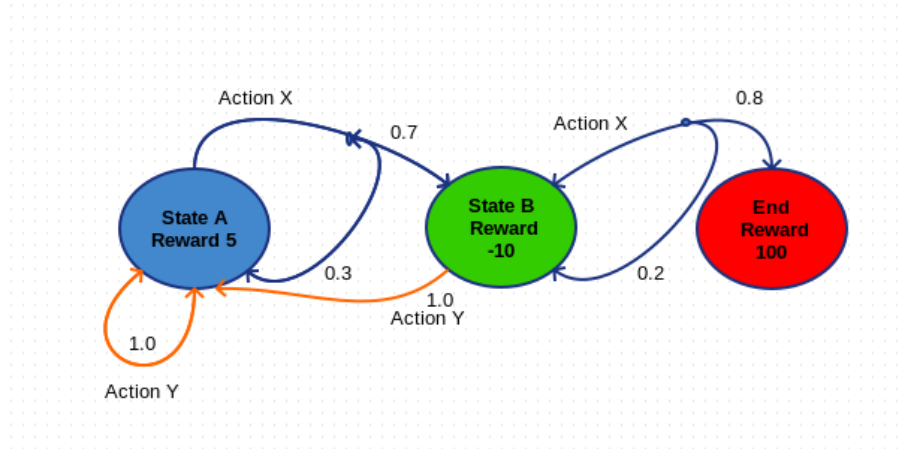


Figure 3.2: A basic MDP [2]

Dynamic Programming and Bellman Equations

Dynamic programming (DP) is a systematic technique for finding optimal policies of an MDP through solving the Bellman equation. The Bellman equation expresses the optimal value of being in a state as the immediate reward available plus the discounted optimal value of the next state. Solving the Bellman Equation requires full knowledge of the environment and state transitions, this is called a model-based approach. The Bellman equation can be defined as [16]:

$$V^*(s) = \max_a [R(s, a) + \gamma V^*(s')] \quad (3.3)$$

where:

- $V^*(s)$ is the optimal value of being in state s
- a is the action taken

- $R(s, a)$ is the immediate reward from taking action a in state s
- $\gamma \in [0, 1]$ is the discount factor, weighting future rewards relative to immediate rewards
- s' is the next state resulting from action a
- $V^*(s')$ is the optimal value of the next state s'

Training Overview - Model Based

Model based reinforcement learning is developed as a Markov decision process with complete knowledge of the environment. A function G is defined to denote the discounted return, and is defined as the sum of future discounted rewards:

$$G = \sum_{t=0}^{\infty} \gamma^t R_{t+1} = R_1 + \gamma R_2 + \gamma^2 R_3 + \dots, \quad (3.4)$$

where:

- R_{t+1} is the reward for transitioning from state S_t to S_{t+1}
- $0 \leq \gamma < 1$ is the discount rate.

Since $\gamma < 1$, rewards in the distant future are weighted less than rewards in the immediate future. The goal of the algorithm is to find a policy that will maximise the expected value of G (π^*). This is achieved through [17]:

- **Policy Evaluation:** Applies the Bellman equation iteratively across all states until the value function converges to a steady value. Beginning from an arbitrary initialisation, each sweep updates the value of every state using the current estimates of neighbouring states.
- **Policy Improvement:** Constructs a new policy from the current value function. At every state, the greedy action is chosen to maximise G . The Policy Improvement Theorem guarantees the new policy is at least as good as the old one for all states.
- **Generalised Policy Iteration:** Alternates between these two steps, evaluating the current policy and improving it. This continues until the value functions converge and the policy is optimal.

Monte Carlo Methods

Monte Carlo methods are based on a model-free approach to RL where there is not complete knowledge of the environment. Monte Carlo methods learn from the experience of interacting

with the environment. They sample and average returns from each state-action pair. Monte Carlo methods use an adapted generalised policy iteration where instead of computing value functions, value functions are learnt. As more returns are observed, the average converges to the expected value. These returns are only used after the agent arrives at the terminal state. [17].

Temporal-Difference

Temporal-difference (TD) learning combines aspects of Monte Carlo and DP. Like Monte Carlo methods, TD works without a model of the environment through learning from the agent's interactions. Similar to DP, TD methods update estimates during state transitions before reaching the terminal state in a process called bootstrapping. Bootstrapping can greatly increase the speed of learning, particularly in environments with long episodes. [17]

Sarsa: On-Policy TD Control

In on-policy TD control, the first step is to learn an action-value function rather than a state-value function. We define the action-value function as $q_\pi(s, a)$ for the current behaviour policy π and for all states s and actions a . Action-value pairs are updated after every transition from a non-terminal state S_t as defined in Equation 3.5. Whilst estimating the state-action pairs, we continually update the policy π towards greediness with respect to q_π . [17]

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha [R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)] \quad (3.5)$$

Q-Learning: Off-Policy TD Control

In Q-learning, the learned action-value function, Q , approximates the optimal action-value function independently of the policy being followed as defined in Equation 3.6. [17]

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha [R_{t+1} + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t)] \quad (3.6)$$

The policy will determine which state-action pairs are visited and updated however the update target always uses $\max_a Q(S_{t+1}, a)$, meaning Q-learning converges to the optimal action-value function regardless of the policy being followed. [17]

Deep Q-Learning

Deep Q-Networks (DQN) combines Q-learning and neural networks. It additionally introduces experience replay and target networks. A visual representation of a DQN is shown in

Figure 3.3 and its features can be summarised as:

- The network approximates the Q-value function $Q(s, a : \theta)$ where θ represents the trainable parameters.
- A replay buffer stores past experiences in the DQN (s, a, r, s') .
- A target network with parameters θ^- is used but updated less frequently to prevent instability in parameters shifting too quickly.
- A loss function, Equation 3.7, that measures the difference between predicted and target Q-Values.

$$L(\theta) = \mathbb{E} \left[\left(R_{t+1} + \gamma \max_{a'} Q(S_{t+1}, a'; \theta^-) - Q(S_t, A_t; \theta) \right)^2 \right] \quad (3.7)$$

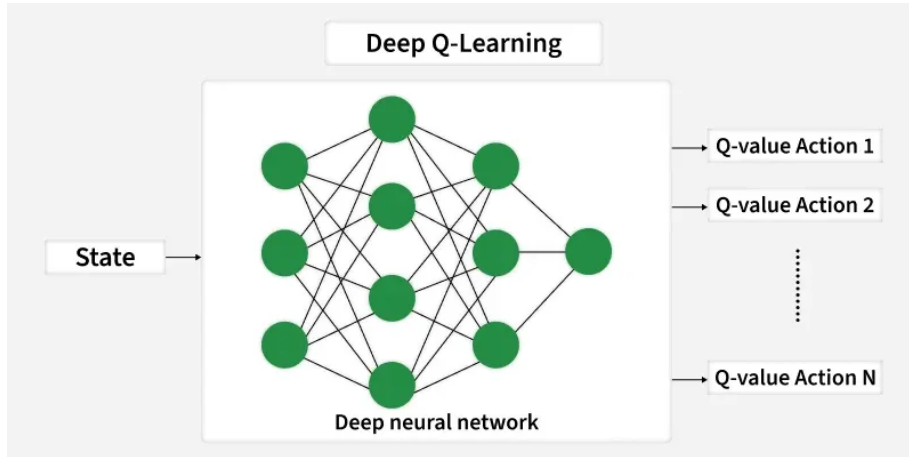


Figure 3.3: Deep Q-Network [3]

3.2 Mixed Integer Linear Programming

Linear programming (LP) involves the optimisation of a linear objective function subject to equality and inequality constraints [18]. This can be expressed generally as:

Find a vector	$\mathbf{x}$
that maximizes	$\mathbf{c}^\top \mathbf{x}$
subject to	$A\mathbf{x} \leq \mathbf{b}$
and	$\mathbf{x} \geq \mathbf{0}$

Mixed integer linear programming (MILP) extends this optimisation but restricts some or all of the variables to take integer values [19]. This presents an alternative to RL for optimising the operation of the household battery.

3.2.1 Branch and Bound Algorithms

Branch and bound algorithms are commonly used by solvers to discover the optimal solution for integer linear programs. The algorithm involves three steps [20]:

1. **Bounding:** At any node on the tree, the problem is relaxed to allow for non-integer solutions. This solution acts as the upper bound (for maximisation problems) of the objective value that can be achieved.
2. **Branching:** From the relaxed solution two branches are created. A fractional variable x_j is selected from the relaxed solution x^* , where x_j^* denotes its relaxed value. One branch adds the constraint $x_j \geq \lceil x_j^* \rceil$ and the other adds $x_j \leq \lfloor x_j^* \rfloor$. The relaxed LP is then solved again.
3. **Pruning:** Pruning involves the removal of branches from the algorithm. There are three types of pruning in branch and bound algorithms.
 - Bound pruning: if the relaxed solution found at a node is worse than the best integer solution found, the branch is pruned.
 - Infeasibility pruning: if the relaxation of the subproblem is not feasible, due to the branching constraints rendering the feasible region empty, the branch is pruned.
 - Integer pruning: if the relaxed solution produces an integer value, update the incumbent value and prune.

3.3 Battery Degradation Mechanisms

Non operational and operational factors main effects

3.4 The Rainflow Cycle Counting Algorithm

How rainflow cycle counting works

3.5 System Characteristics

3.5.1 Network Tariffs

Network tariffs are charged by electricity distributors to retailers for delivering electricity. The charge is associated with the infrastructure costs, maintenance, and operation of electricity transmission. Network infrastructure in NSW is a regulated asset, owned by private companies that are entitled to fixed revenue set by the Australian Energy Regulator [21]. This project will consider low-voltage residential connections on the Ausgrid network. This tariff consists of a daily metering service and network access charge along with a network energy price of 10.8007 cents/kWh [22].

3.5.2 Battery and Inverter Models

There is a wide variety of residential batteries available in the Australian market. Under the Small-scale Renewable Energy Scheme introduced in July 2025, the average nominal battery size has been 19.0 kWh [23]. This project does not model a particular product. It considers a generic lithium-ion battery whose electrical and aging parameters follow the test battery of Xu et al. [11], at a capacity and inverter rating typical of a single residential unit. The parameters are summarised in Table 3.1. The state of charge is held between 15% and 95% of the rated capacity, so 10.8 kWh of the 13.5 kWh is usable.

Table 3.1: Battery parameters

Parameter	Value	Source
<i>Energy & Power</i>		
Rated energy capacity E^{rate}	13.5 kWh	Residential scale
Max. charge power P_{max}^c	5 kW _{AC}	Residential scale
Max. discharge power P_{max}^d	11.04 kW _{AC}	Residential scale
<i>Operation</i>		
Charging and discharging efficiency η^c, η^d	95%	[11]
Maximum state of charge E_{max}	95% (12.825 kWh)	[11]
Minimum state of charge E_{min}	15% (2.025 kWh)	[11]
<i>Aging</i>		
Cell chemistry	Li(NiMnCo)O ₂	[11]
Cycle life	3000 cycles at 80% cycle depth	[11]
Shelf life	10 years	[11]
Cell temperature	25 °C	[11]
Replacement cost R^{cell}	\$800/kWh (\$10 800)	[24]

Xu et al. use a pack replacement cost of US\$300/kWh for a grid-scale battery. A household pays considerably more per unit of capacity, so the replacement cost is instead

set from Australian installed prices: about \$750 to \$830/kWh for 10 to 20 kWh systems including installation and GST, before the federal rebate [24]. A figure of \$800/kWh of rated capacity is adopted. Because an installed price includes labour and balance of system as well as the cells, it is an upper estimate of the cost of replacing them.

3.5.3 Cycle Depth Stress Function

The degradation model of Section 2.2.2 requires a cycle depth stress function $\Phi(\delta)$ and a replacement cost R^{cell} to evaluate the marginal aging costs c_j in (2.3). Xu et al. [11] fit a near-quadratic stress function to Li(NiMnCo)O₂ 18650 cells:

$$\Phi(\delta) = (5.24 \times 10^{-4}) \delta^{2.03} \quad (3.8)$$

The exponent greater than unity is what makes the marginal cost curve convex, and hence what makes deep cycling progressively more expensive per unit of energy than shallow cycling.

Equation (3.8) is consistent with the cycle life in Table 3.1: $\Phi(0.8) = 3.33 \times 10^{-4}$, or one three-thousandth of cell life per cycle of 80% depth. Cycle depth δ is measured as a fraction of the rated capacity, so the state of charge limits cap the deepest possible cycle at 80%. The stress function is fitted to NMC chemistry. Lithium iron phosphate (LFP) cells exhibit a flatter relationship between cycle depth and life, so (3.8) would overstate the penalty on deep cycling for an LFP battery.

3.5.4 Solar

The average solar installation size has continued to increase. In the first quarter of 2025, the average installation size in NSW was 9.9 kW [25].

Chapter 4

Current State of Literature

This chapter will explore the key areas of literature relevant to this project. Most literature separately studies the modelling of price forecasting, load forecasting, battery characteristics and trading optimisation. Each of these areas will be discussed individually.

4.1 State Forecasting

4.1.1 Electricity Prices

The electricity wholesale price in the NEM experiences frequent price fluctuations driven by supply-demand imbalances [9]. Statistical modelling techniques have traditionally been used to predict pricing and seasonal variations. Alharbi and Csala [26] demonstrated the effectiveness of Seasonal Autoregressive Integrated Moving Average with Exogenous Factors (SARIMAX) forecasting for electricity prices in Saudi Arabia.

Zhou et al. [27] compared a variety of prediction models on the NSW NEM. Three distinct time frames were used to enhance accuracy and robustness of forecasting. A time frame with significant volatility, limited volatility and significant but temporary volatility were selected. The features selected to inform the prediction were the Regional Reference Price (RRP), FCAS prices and system demand.

The effectiveness of the models were evaluated using metric including Mean Absolute Error (MAE), Root Mean Squared Error ($RMSE$), Mean Absolute Percentage Error ($MAPE$), and R-squared (R^2). It was shown that XGBoost achieved the most accurate forecasts. LSTM and SARIMAX were shown to provide reasonable results in stable regimes but with declining accuracy over periods of volatility.

4.1.2 Solar Output

The most important factor in determining output from a residential solar system is weather conditions. However, meteorological forecasting has considerable uncertainty that must be considered when predicting future solar generation. Zhong et al. [28] systematically compared Numerical Weather Prediction (NWP) input variables across ANN, LSTM, and XGBoost models. The study found that XGBoost with all NWP variables, cumulative irradiation, and cumulative rainfall achieves the lowest forecasting error, outperforming both ANN and LSTM. The study further suggests that for a model with only irradiance data available, XGBoost continues to outperform ANN and LSTM models.

4.1.3 Residential Load

Expected future residential load plays a significant role in deciding if a home should charge or discharge a battery. Cao et al. [29] presented an autoregressive integrated moving average (ARIMA) model and Similar Day Method for load forecasting. It demonstrated that ARIMA worked best for stable weather days whilst Similar Day Method outperformed ARIMA on days with sudden weather shifts.

Machine learning has been used for residential load prediction. Kong et al. [4] used machine learning to forecast residential loads in NSW and found LSTM outperformed backpropagation neural networks (BPNN) and extreme learning machines (ELM). The paper found that aggregating the loads at the substation level improved the accuracy of the methods, indicating the difficulty in dealing with highly variable individual household loads.

4.2 Electricity Trading

4.2.1 Commercial Battery Energy Storage Systems

The majority of research in battery arbitrage has been conducted on commercial scale battery energy storage systems (BESS). Karimi-Arpanahi et al. [30] outlined a MILP methodology for battery deployment in the NEM for both wholesale and FCAS markets. The study compares decisions made by the MILP model to actions by battery operators and highlights the unrealised revenue based on the existing BESS bidding strategies. The paper highlighted that price forecast inaccuracy was a heavy indicator in simulated BESS underperforming.

Cao et al. [12] outlined a machine learning model for predicting and trading wholesale electricity on the UK market. It identified three simplifying assumptions made in existing

literature as: 1) perfect foresight about electricity market prices; 2) constant battery charging/discharging efficiency; 3) simple representation of battery degradation. The MDP problem formulation for energy arbitrage outlined by Cao et al. is defined as:

1. State space: The state space at time instant t is defined as $s_t = (c_t, \dots, ct_{22}, ct_{23}, SoC_t)$. where $c_t, \dots, ct_{22}, ct_{23}$ is the predicted price for the next day. Using the predicted price signal is to make sure the agent knows whether the price signal is going up or down in order to make the best control action.
2. Action space: The charging/discharging action space is discretised as

$$a = (-P_e^{\max}, -0.5P_e^{\max}, 0, 0.5P_e^{\max}, P_e^{\max}) \quad (4.1)$$

where $P_e^{\max}$ is the maximum charging/discharging power of the battery inverter.

3. Reward: The design of reward function is the key factor in the algorithm. The reward in the energy arbitrage problem should include not only the profit from the discharging action, but also the degradation costs of the control action.

The paper compares reinforcement learning models and finds that NoisyNet-DoubleDDQN (NN-DDQN) models outperform the proposed MILP method by 58.51%. It also found that NN-DQN models outperformed Vanilla DQN.

4.2.2 Residential Battery Energy Storage Systems

Scheduling batteries in home energy management systems requires the consideration of PV output, load demand, battery degradation costs and wholesale pricing. Correra et al. [31] designed a MILP solution for households in Portugal with PV, BESS and electric water heater (EWH) systems to minimise electricity costs. The paper demonstrated that a stochastic optimisation of the problem outperformed a deterministic optimisation due to the variability of PV and load forecasting. The model used a simplified persistence model for day-ahead price forecasting that would not be suitable for the variability shown in the NEM.

Abdulla et al. [32] proposes a stochastic dynamic programming (SDP) approach to operating BESS systems for households in Australia. The paper outlined three degradation models of increasing fidelity from fixed per kWh, static multi-factor and dynamic multi-factor degradation. The paper finds that when considering battery lifetime value, the SDP considering degradation achieves a net return of +2.3% whilst when battery degradation is ignored a net return of -3.1% is achieved.

4.2.3 Residential Demand Response

An opportunity for residential consumers to respond to market pricing signals is through demand response (DR). DR involves customers adjusting their electricity consumption based on pricing signals. In [33], the authors analysed the potential for demand management in response to negative wholesale prices in the NEM. It found that the number of negative price events with DR opportunities are increasing with greater variable renewable energy penetration. Mahmoudi et al. modelled DR for energy retailers in the NEM. It was found that through aggregating DR across the customer base, retailers were able to achieve a reduction in wholesale electricity purchase volume during peak pricing events [34]. Farzambehboudi et al. [35] conducted modelling on a household with smart metering data in the Turkish electricity market. They concluded that load shifting would reduce household electricity expenditure by 20% on a Time of Use (ToU) pricing structure through controlling existing household appliances. In [36], thermal storage air-conditioning was modelled with DR to wholesale prices in the NEM. The paper found that the demand response program not only reduced the purchase cost of electricity for the program owner but also reduced the market clearing price for the NEM. There are additional significant opportunities for demand response from EV charging. Through adjusting EV charging based on market signals, significant consumer savings and demand reduction can be achieved [37].

4.3 Summary

There is a large amount of research into the optimisation of electricity generation and trading. The modelling of price, solar and load are well researched areas that have direct economic consequences. While research has been conducted for consumer energy resources, it often focuses on simplified models with consistent electricity prices. The papers also often make simplifying assumptions regarding battery degradation and forecasting. This thesis fills a gap by analysing wholesale electricity prices using consumer energy resources with accurate models for degradation and forecasting.

Chapter 5

Data Analysis

5.1 National Electricity Market

Data from the National Electricity Market was studied to identify trends and features that could be used in the forecaster model.

5.1.1 Data Structure

NEM data was collated for the period from January 2015 to December 2024 for the NSW NEM region. It contains settlement date-time, total demand for the NEM region, and wholesale price per MWh as determined by AEMO. The settlement periods are in 30 minute intervals until the 1st of October 2021, followed by 5 minute settlement periods for the rest of the data.

5.1.2 Demand

Figure 5.1 shows the demand in the NSW NEM over the 10 years with a 30 day rolling average. Figure 5.2a shows the daily trends of demand whilst Figure 5.2b shows the seasonal variation in demand.

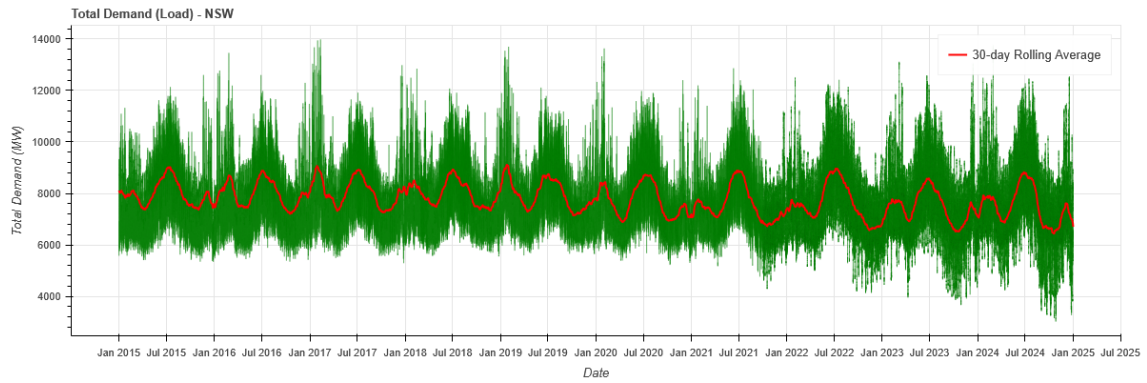
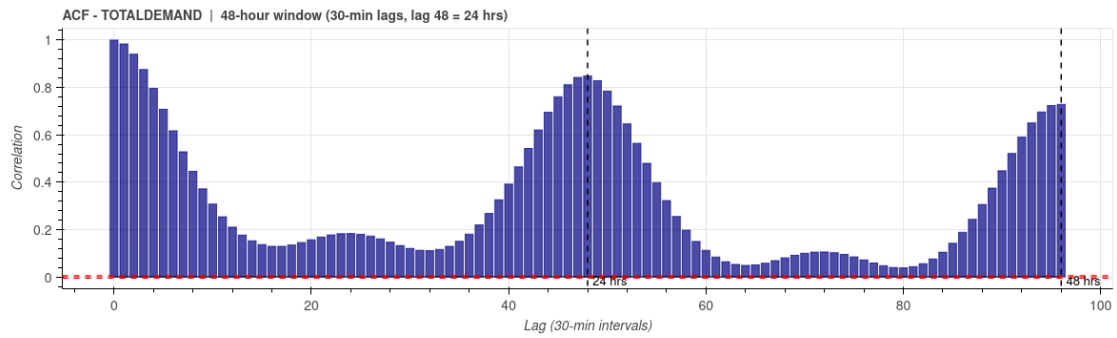
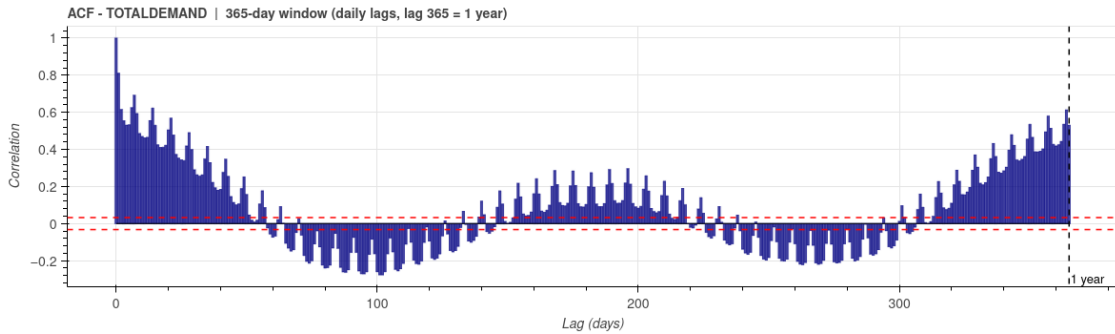


Figure 5.1: Demand in NSW NEM with 30 day rolling average



(a) ACF - 48 Hours



(b) ACF - 365 Days

Figure 5.2: Autocorrelation analysis of demand in NSW NEM

5.1.3 Regional Reference Price

The data analysis on the regional reference price (RRP) found a less pronounced autocorrelation than on demand. Figure 5.3 shows the wholesale price in the NSW NEM with a rolling average. The figure shows an increasing number of price spike events occurring. Figure 5.4 has been normalised to remove the 1st and 99th percentiles. The results do not show the same seasonality as demand. Figure 5.5 however indicates that whilst long-term trends are difficult to interpret in price data, there is a strong daily trend shown by the peaks at 24 and 48 hours.

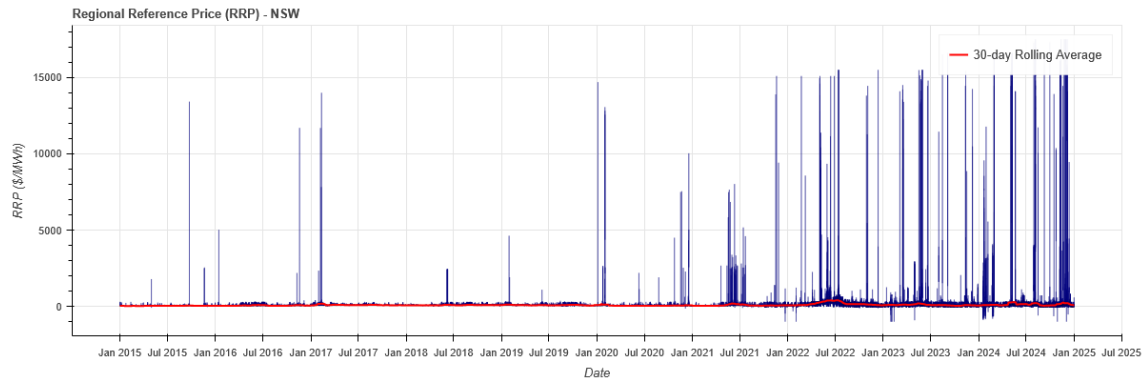


Figure 5.3: Wholesale price in NSW NEM with 30 day rolling average

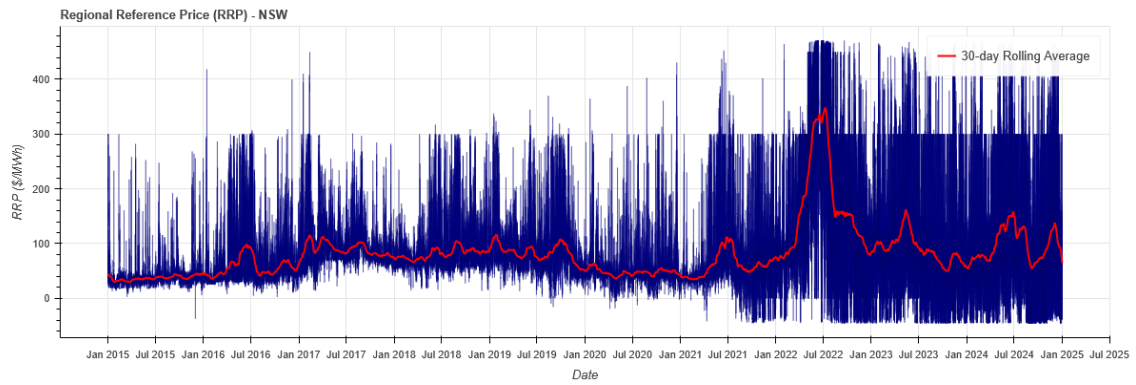


Figure 5.4: Wholesale price in NSW NEM with 1st and 99th percentile removed

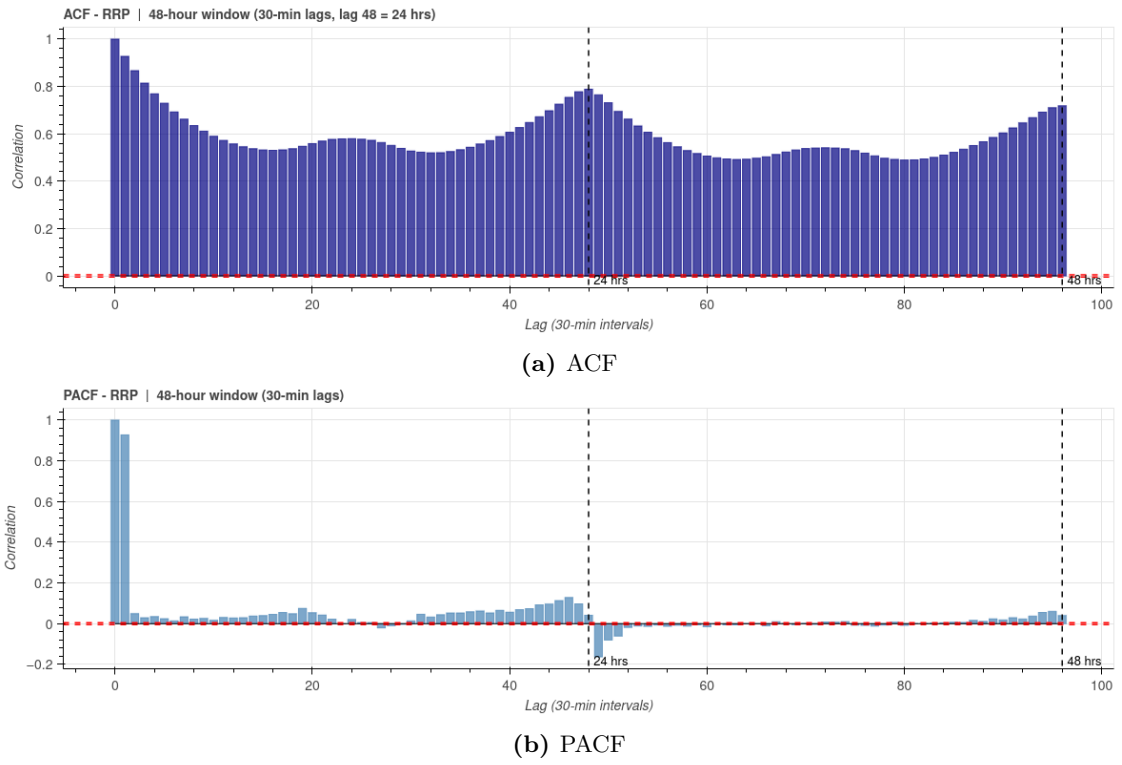


Figure 5.5: Autocorrelation analysis of normalised NSW NEM price series

5.2 Net Local Power

Chapter 6

Implementation

6.1 MILP Optimisation

Mixed-integer linear programming was used to control the operation of the battery. The system model and constraints outlined in Chapter 2 were implemented into the PuLP Python package. The system was then solved using HiGHS optimisation software. The optimiser plans over a 24-hour horizon using the forecast price and load along with the data for the current interval. The simulation then commits only the first intervals charge and discharge actions and then recomputes over the next 24-hour horizon. The MILP optimiser runs with a variety of forecasting methods for price and load.

6.2 Battery Degradation

Battery degradation is represented inside the MILP objective function by the J-segment cycle aging cost which provides a linear approximation for battery degradation. Different values for J were studied to determine the system behaviour and accuracy of the piece-wise linearisation. After the optimisation completed, rainflow cycle counting was used. The rainflow counting algorithm has been applied to battery life assessment to determine the number of equivalent cycles of battery discharges [38] [39]. The rainflow algorithms is used as a benchmark method for assessing battery cycle life as in Xu et. al [11]. The total life lost L from a SoC profile is assumed to be the sum of the life loss from all I number of cycles. Using the cycle depth stress function $\phi(\delta)$ given a cycle depth of δ , the life loss in polynomial form is:

$$L = \sum_{i=1}^I \Phi(\delta_i). \quad (6.1)$$

The benchmark is then used for comparing different piece-wise approximations and

forecasting models.

6.3 LSTM Models

LSTM models were used to forecast both wholesale price and the net local power. Both models were implemented in PyTorch and share a common structure. A stacked LSTM layers read a window of past state values, and the final hidden state of the last layer is passed through a small network that outputs the forecasted values of the 24 hour horizon. A forecast is issued at the start of each step using data up to the end of the previous step. Each forecasted value is held constant across the time-slot.

6.3.1 Price Forecaster

The price model takes the values of three features over the previous seven days as input. The first feature is the regional price transformed as $\sinh^{-1}(R/100)$. The transformation is approximately linear near zero and logarithmic for large prices. This prevents the large price peaks from skewing the weights of the model. The second feature is regional demand, standardised by its mean and standard deviation over the training period. The third feature is six calendar features: the sine and cosine of the time of day, day of week and day of year. The network is a two-layer LSTM with 64 hidden units and a drop of 0.2. Forecasts are returned to \$/Mwh from the final hidden state by $100 \sinh(\hat{x})$.

The model was trained on NEM data from 2015 to 2023 and validated on 2024 data. The five-minute data after October 2021 was averaged to half-hours to maintain consistent resolution. The Huber loss function was used on the transformed price output which limited the influence of large errors caused by unpredicted price peaks. The optimiser was Adam with an initial learning rate of 10^{-3} and a batch size of 256. The learning rate was halved after two epochs without improvement over the validation data. Training stopped early after five epochs without improvement to prevent overfitting, and the weights from the epoch with the lowest validation loss were used.

6.3.2 Net Local Power Forecaster

The net local power model follows the residential load LSTM in Kong et al. [4]. Each slot has four inputs. The first input is the net local power, $G - A$, min-max scaled over the range of training data. The other inputs are the half-hour of the day, the day of the week and a flag for NSW public holidays. As in Kong et al., the network is two stacked LSTM layers of 20 units. The final hidden state passes through a fully connected layer of 64 units with ReLU activation, then a final linear layer with 48 outputs. The model was trained

with a mean squared error loss, using Adam with a learning rate of 10^{-3} and a batch size of 64.

The Kong et al. model was adjusted to have an output layer with 48 outputs instead of one. This reflected the forecasters requirement to predict over a 24-hour period instead of a 30-minute period. Additionally, the look-back was set to one day (48 half-hour periods), against the 2 to 12 half-hours used by Kong et al. The model was trained on data from January to October 2024, validated on November and December 2024, and tested on January 2025.

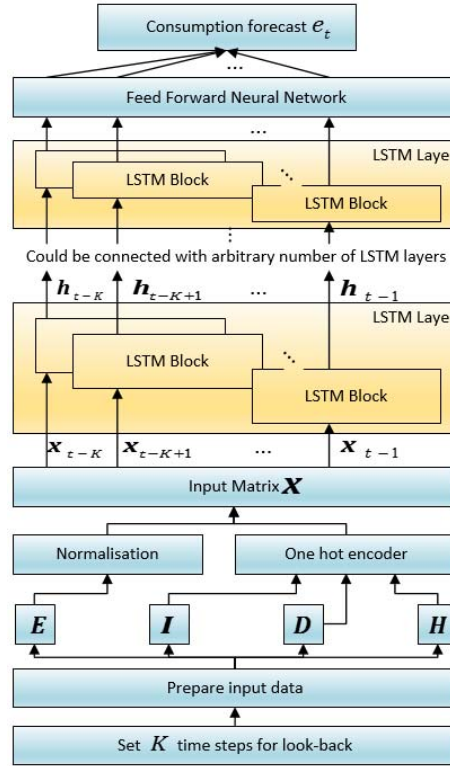


Figure 6.1: LSTM forecasting framework [4]

6.4 Additional Forecasters

6.4.1 Seasonal Naive

This is the simplest forecaster considered. The forecaster predicts that the price over an interval is equal to the price over the same interval the day before.

$$\hat{R}_{k+j|k} = \bar{R}_{k+j-48}, \quad j = 0, 1, \dots, 47 \quad (6.2)$$

6.4.2 AEMO Pre-Dispatch

The Australian Energy Market Operator publishes price forecasts for each NEM region. The forecasts are recalculated every half-hour and provide a half-hourly price for every period to the end of the following market day. AEMO calculated the price forecasting by considering; bids submitted by participants, AEMO's demand forecast, AEMO's intermittent generation forecasts and constraints including network condition, FCAS requirements and inter-connector limits [40]. This forecasting method is used in Karimi-Arpanahi et al. [30].

6.4.3 7-day Load Profile

Chapter 7

Results

7.1 State Machine

To validate the basic operation of a Python framework that consumed NEM and household data, orchestrated battery charge/discharge and produced results for household cost, a basic state machine was designed as shown in Figure 7.1.

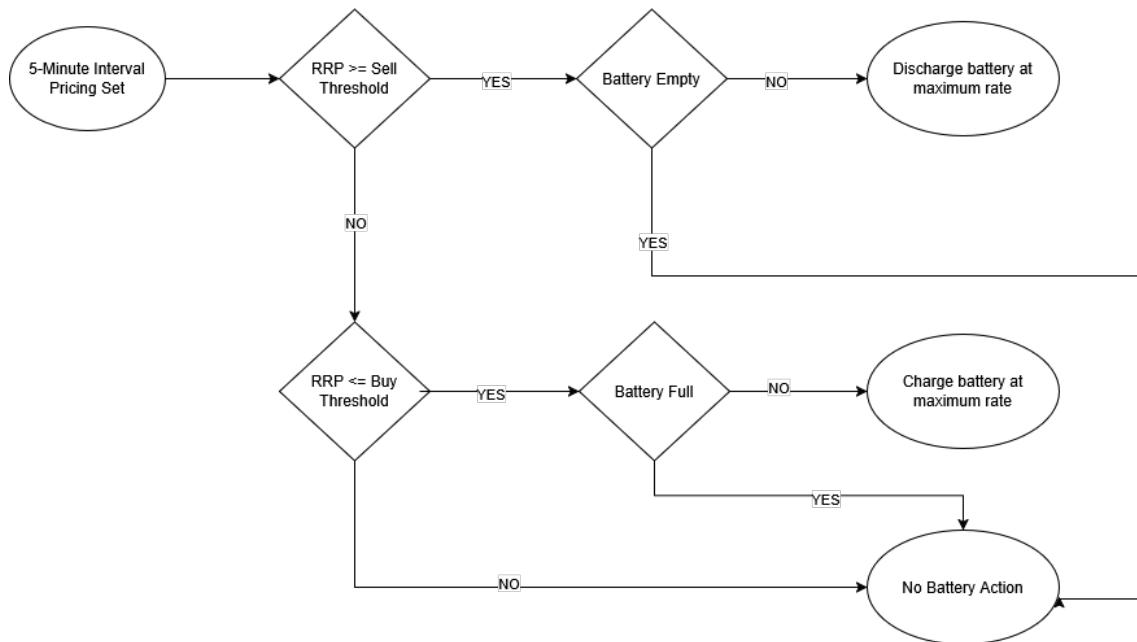


Figure 7.1: A flow chart for state machine trading

The model was trained on a grid search to determine sell thresholds and buy thresholds for a lossless 20 kWh battery with an 11.04 kW inverter, which starts the month empty. Surplus solar may charge the battery at up to 20 kW. A 60x60 grid search was used with buy ranges from \$0/MWh to \$87.50/MWh and a sell range from \$87.50 to \$189.33. The grid value that produced the lowest net-cost to the consumer was selected. The grid was

searched using data from December 2024 NEM RRP and household import/export data. Dispatch intervals with an RRP outside the 15th to 85th percentile range were removed before the search, which leaves 6256 of the 8928 intervals, and the battery was simulated across the remaining intervals as one continuous series. The net profit reported for the search is therefore the result on that reduced series and not the result for the whole of December. The result of the optimisation is summarised in Table 7.1.

Parameter	Value
Buy threshold	\$69.70
Sell threshold	\$127.20
Net profit on the reduced Dec. 2024 series	\$-34.35

Table 7.1: Optimised Thresholds

The state machine was then simulated on data from January 2025 with the thresholds in Table 7.1. The results are summarised in Table 7.2. Figure 7.2 shows the SoC of the battery in relation to RRP and excess solar production. The battery follows a pattern of charging during the day from excess solar and low RRP and discharging in the afternoon.

Parameter	Value
Final Battery State	18.16 kWh
Total Grid Cost	\$126.34
Total Grid Revenue	\$179.50
Net Profit	\$53.16

Table 7.2: State Machine Results January 2025

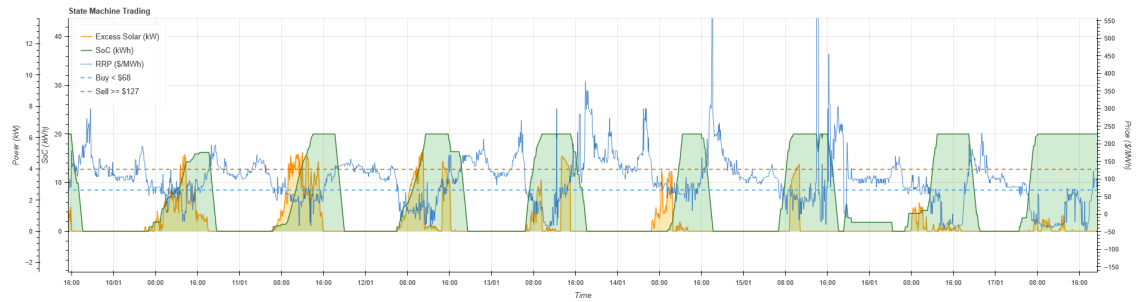


Figure 7.2: One week period of state machine control January 2025

Figure 7.3 provides insight into the pricing events that are contributing most to the export revenue generated. Figure 7.3b shows that exports during the top 0.1% of dispatch intervals by price, the nine intervals at or above \$1510/MWh, contribute 52.5% of total export revenue over the month. The battery orchestration exported during all nine of these

intervals. This distribution of revenue highlights the importance of capturing peak pricing events. It additionally raises questions about how to future-proof a trading system where price peaks become less common as greater commercial BESS capacity is brought online.

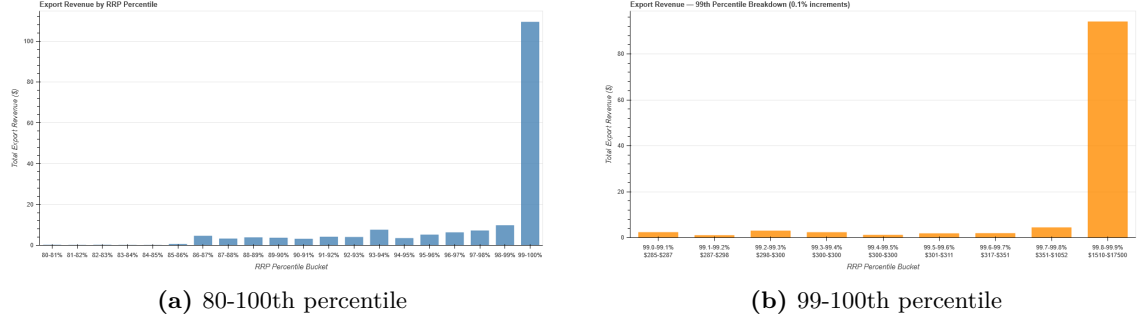


Figure 7.3: Export revenue by dispatch period RRP percentile

7.2 Genie Model

Establishes an upper bound of to measure the effect of cycling age cost

7.2.1 Upper Bound Results

Result that sets the upper bound for our further study

7.2.2 Battery Degredation Model

Ignoring degradation removes the value of trading.

The MILP was solved with a rolling horizon: each window covers 48 hours, the first 24 hours are committed, and the per-segment energy is carried into the next window, giving 31 windows for the month. Two scenarios were run. In the *household* scenario the battery sits behind the meter with the solar and load. In the *battery-only* scenario the household terms are zero and the battery trades alone. Results are given in Tables 7.3 and 7.4.

Ignoring degradation removes the value of trading. With $R^{\text{cell}} = 0$ the optimiser earns the most at the meter, \$95.04 in the household scenario and \$136.23 for the battery alone, but it completes 42 and 25 equivalent full cycles in the month and consumes 1.6% and 1.2% of the battery’s cycle life. Once that wear is valued by rainflow counting, both scenarios make a net loss. The threshold state machine fails in the same way: it earns more at the meter than any degradation-aware MILP and loses all of it to aging, for a net result of $-\$83.53$.

Table 7.3: Perfect-foresight MILP, household scenario, January 2025. Error is the relative difference between the degradation cost inside the MILP and the rainflow cost. The state machine uses thresholds of \$69.70 and \$127.20 per MWh on a 20 kWh lossless battery and is included for qualitative comparison only.

Model	Profit at meter (\$)	Rainflow deg. (\$)	Net profit (\$)	Life lost (%)	Full cycles	Error (%)
$J = 1, R^{\text{cell}} = 0$	95.04	194.62	-99.58	1.622	42.4	-
$J = 1$	14.40	2.40	12.00	0.020	1.0	170
$J = 2$	43.52	22.61	20.91	0.188	8.0	15.4
$J = 4$	43.31	15.07	28.25	0.126	9.3	4.1
$J = 8$	47.69	16.86	30.83	0.141	12.7	0.9
$J = 16$	48.62	16.67	31.95	0.139	15.0	3.9
State machine	53.16	136.70	-83.53	1.139	25.6	-

Table 7.4: Perfect-foresight MILP, battery-only scenario, January 2025.

Model	Profit at meter (\$)	Rainflow deg. (\$)	Net profit (\$)	Full cycles	Error (%)
$J = 1, R^{\text{cell}} = 0$	136.23	139.31	-3.09	25.4	-
$J = 1$	103.98	1.68	102.30	0.9	252
$J = 2$	105.91	2.94	102.98	1.4	51
$J = 4$	113.97	6.77	107.20	4.8	12.5
$J = 8$	113.41	5.22	108.19	5.0	6.9
$J = 16$	113.55	5.16	108.39	5.4	2.6

A degradation-aware controller trades about a fifth as often and keeps the profit.

At $J = 4$ the household battery completes 9.3 equivalent full cycles, loses 0.126% of its life, and returns a net profit of \$28.25. Relative to the no-battery position this is \$120.04 of value added after \$15.07 of aging.

7.3 Price Forecasters

7.3.1 Seasonal Naive

7.3.2 AEMO Pre-Dispatch

7.3.3 LSTM

7.4 Net-Demand Forecasters

7.4.1 7-Day Load Profile

7.4.2 LSTM

7.5 Spot trading and retail comparison

This chapter reports three studies carried out on the battery model of Chapter 2. The first solves the mixed-integer linear program (MILP) with perfect foresight to establish an upper bound and to measure the effect of the cycle-aging cost. The second replaces perfect foresight with forecasts inside a model predictive control (MPC) loop, and measures how much of that upper bound a realisable controller recovers. The third operates the same battery for self-consumption on a set-rate retail tariff, as a reference for what a household would obtain without market exposure.

7.5.1 Common setup

All results are for the 31 days of January 2025 in the NSW1 region, which is 8928 five-minute dispatch intervals. All monetary values are in Australian dollars for that month.

The battery is the generic lithium-ion battery of Table 3.1, with a rated capacity of 13.5 kWh operated between 15% and 95% state of charge, a maximum charge power of 5 kW, a maximum discharge power of 11.04 kW, charge and discharge efficiencies of $\eta_c = \eta_d = 0.95$, and an initial energy of $E_0 = 6.75$ kWh. The household is a single dwelling with rooftop solar. Its meter records import and export only, so the model observes the net local power $G_n - A_n$ and never G_n and A_n separately. The meter registers both an import and an export in 636 of the intervals. The two channels are netted within each interval in every study in this chapter, including the no-battery references, because the dispatch models see only the net. On that basis the household without a battery imports 458 kWh and exports 490 kWh over the month (464 kWh and 495 kWh on the raw channels).

Under spot settlement, imported energy is charged at the spot price plus the network tariff, $R_n + N$ with $N = 10.80$ c/kWh [22], and exported energy is paid the spot price R_n .

Battery degradation is represented inside the MILP by the J -segment cycle-aging cost of Xu et al. [11]. Every dispatch schedule is additionally valued after the fact by rainflow cycle counting, using the stress function $\Phi(\delta) = 5.24 \times 10^{-4} \delta^{2.03}$ and a cell replacement cost of $R^{\text{cell}} = \$10\,800$ (\$800/kWh). Throughout this chapter, *net profit* means profit at the meter less the rainflow degradation cost, so that schedules produced with different values of J are valued on the same basis.

The spot price in January 2025 had a mean of \$83/MWh and a median of \$76/MWh. It was negative in 15.2% of intervals, above \$300/MWh in 0.45% of intervals, and above \$1000/MWh in ten intervals, which fell on 1, 14, 15 and 22 January. The maximum was the market price cap of \$17 500/MWh on 15 January.

Four segments are sufficient. With a single segment, all cycling is priced at one marginal cost and the battery barely trades. Net profit rises steeply up to $J = 4$ and then flattens: \$28.25, \$30.83 and \$31.95 at $J = 4, 8$ and 16 , for solve times of 12 s, 27 s and 59 s. At $J = 4$ the degradation cost inside the MILP agrees with the rainflow cost to within 4.1%. The error is not monotonic in J beyond this point (0.9% at $J = 8$ and 3.9% at $J = 16$). The error is an absolute value and its sign changes: the MILP overstates the rainflow cost at $J = 4$ and $J = 8$ and understates it at $J = 16$. Below about 5% the error is therefore no longer controlled by the number of segments. The cause has not been isolated; the candidates are the 24-hour commitment of a 48-hour window, the terminal constraint applied to every window, and the solver optimality gap. $J = 4$ is used for the forecasting study as the smallest value at which the error is below 5%.

Co-located solar and load increase the value of the battery. Before aging, the battery adds \$135.11 in the household scenario against \$113.97 when trading alone. This is consistent with the battery charging from surplus solar that would otherwise be exported at R_n , and discharging into load that would otherwise be imported at $R_n + N$, so that it earns the network tariff in addition to the price spread.

7.5.2 Forecast-driven dispatch

This study measures the *dispatch value* of a forecast: the net profit obtained when the MILP plans on that forecast, compared with the net profit under perfect foresight. The household scenario is used with $J = 4$.

At every five-minute interval the controller solves the MILP over a 24-hour horizon, implements the first interval only, and repeats, giving 8928 solves per forecaster at 0.14 to 0.33 s each. Following the information structure of Chapter 2, the price of the current

interval is known, because AEMO publishes it before the interval begins, while all later prices and all net local power, including that of the current interval, are forecast. The implemented charge and discharge powers are then settled against the actual price and the actual net local power.

Three price forecasters are compared. Each issues a forecast of the mean price in each of the next 48 half-hours, which is held constant across the six dispatch intervals of the half-hour.

- *Seasonal naive*: the mean price of the same half-hour on the previous day.
- *AEMO pre-dispatch*: the regional price from the most recent AEMO pre-dispatch run published at or before the start of the interval, with intervention runs excluded. A run extends to the end of the next trading day, which can be less than 24 hours ahead, and the remainder of the horizon is filled with the seasonal-naive forecast.
- *LSTM*: a two-layer LSTM with 64 hidden units and a dropout of 0.2 reads the previous seven days (336 half-hours) of price, regional demand and calendar features (sine and cosine of the time of day, day of week and day of year). Its final hidden state is concatenated with the calendar features of the 48 target half-hours and passed through a fully connected layer of 128 units to produce all 48 forecasts at once. Prices are transformed by $\sinh^{-1}(R/100)$, which is approximately linear near zero, logarithmic for large prices and defined for negative prices, and demand is standardised. The network is trained with a Huber loss on the transformed prices using Adam with a learning rate of 10^{-3} and a batch size of 256, on 2015 to 2023, with early stopping on the 2024 validation loss. A forecast is issued at the start of every half-hour from data up to the end of the previous half-hour.

The literature behind these choices is reviewed in Section 4.1.1. The net local power is forecast in every case by the mean of the same five-minute slot over the previous seven days, which has a mean absolute error of 1.15 kW. The *perfect price* case uses actual prices with this household forecast, and so separates the cost of the household forecast from the cost of the price forecast.

On the 2024 validation year the LSTM had an MAE of \$70.3/MWh against \$103.1/MWh for the seasonal-naive forecast, an rMAE of 0.68, with the best model obtained at epoch 5. January 2025 was not used for training or model selection.

The rolling loop costs nothing under perfect information. Perfect foresight on the 24-hour, five-minute MPC loop gives the same net profit as the 48-hour rolling solve to within one cent (\$28.74), although the two schedules differ in 4% of intervals. Every gap in

Table 7.5: Forecast accuracy and dispatch value, household scenario, $J = 4$, January 2025. MAE is the mean absolute error of the price forecast over all lead times and rMAE is the MAE relative to the seasonal-naive forecast.

Forecaster	MAE (\$/MWh)	Median AE	rMAE	Profit at meter (\$)	Rainflow deg. (\$)	Net profit (\$)	% of perfect
Perfect	0	0	–	43.85	15.11	28.74	100
Perfect price	0	0	–	34.33	14.69	19.64	68
LSTM	44.6	22.0	0.73	33.29	17.35	15.94	55
Seasonal naive	61.3	25.3	1.00	31.71	15.79	15.92	55
AEMO pre-dispatch	161.6	16.4	2.63	30.24	15.57	14.67	51

Table 7.5 is therefore caused by forecast error and not by the horizon or the re-planning frequency.

Most of the loss comes from the household forecast, not the price forecast.

Knowing every price but forecasting the net local power loses \$9.10 of the \$12.80 to \$14.07 total gap. Measured against the perfect-price case, the price forecasts cost \$3.70 for the LSTM, \$3.72 for the seasonal-naive forecast and \$4.97 for AEMO pre-dispatch.

A more accurate price forecast did not produce more dispatch value. The LSTM reduces the MAE by 27% relative to the seasonal-naive forecast, from \$61.3/MWh to \$44.6/MWh, and finishes two cents ahead of it. It earns \$1.59 more at the meter, but its schedule has deeper cycles (a mean depth of 0.196 against 0.158) and incurs \$1.56 more in aging.

AEMO pre-dispatch is the most accurate forecast in normal conditions and the least accurate on average. It has the lowest median error (\$16.4/MWh) and the highest MAE (\$161.6/MWh). It forecast prices near the market price cap on the hot days of 14–15 and 21–22 January that largely did not eventuate: 1.0% of its forecast values exceed \$1000/MWh, against 0.1% of actual prices. With every forecast value capped at \$1000/MWh before the error is taken, its MAE is \$48.9/MWh, between the LSTM (\$44.7/MWh) and the seasonal-naive forecast (\$53.2/MWh). Actual prices are not capped in this measure, so a perfect forecast scores \$11.6/MWh on it. In dispatch it finishes \$1.25 below the seasonal-naive forecast.

Forecast accuracy and dispatch value are only loosely related. Ranked by MAE the order is LSTM, seasonal naive, AEMO pre-dispatch. Ranked by net profit the LSTM and the seasonal-naive forecast are level and AEMO pre-dispatch is slightly behind, with all three within \$1.27 of one another.

Implications

The assumption that carries the most weight in this study is that the price of the current interval is known before the dispatch decision is made. It is what secures the \$101.53 earned during price spikes. The assumption is realistic, but its value has not yet been measured, and doing so requires only that the study be repeated with the current-interval price forecast rather than observed.

The spike revenue was captured because the battery was not empty when each spike arrived. One month containing four spike days is not evidence that a controller will generally hold charge at the right time. A reserve state of charge, or a spike-probability input, would address that risk. It would not close the \$12.80 gap measured here, because that gap does not arise from spikes.

The largest recoverable loss is the household forecast, at \$9.10. An improved forecast of net local power is worth more than any further improvement to the price model. Part of this loss is a property of the controller and not of the forecast. The loop commits a battery power for the whole interval from a forecast of the net local power, and the forecast is a seven-day mean that does not use the most recent meter readings, although Chapter 2 makes them available. A battery inverter measures the net load continuously, so a controller that committed a grid power target, or that corrected the forecast with the latest readings, would recover some of the \$9.10 without a better day-ahead forecast.

7.5.3 Retail self-consumption

As a reference for a household without market exposure, the same battery was operated for self-consumption on two published AGL Residential Smart Saver plans for the Ausgrid network area [41], both with a feed-in tariff of 3.0 c/kWh. The single-rate plan charges 29.82 c/kWh with a supply charge of 149.57 c/day. The time-of-use plan charges 54.18 c/kWh from 3 pm to 9 pm and 21.63 c/kWh otherwise, with a supply charge of 158.63 c/day. The retail rates include GST.

Two controllers are compared on each plan. The *self-consumption rule* charges from surplus solar, discharges into household load, and never exchanges energy with the grid on the battery's behalf. On a single-rate tariff with no export limit this rule is optimal before aging, since the import rate exceeds the feed-in rate at every instant. It has no price signal against which to weigh a cycle, so its aging is only valued after the fact by rainflow counting. The rule has no terminal constraint, so it is started from the state of charge that the same rule reaches at the end of December, which is empty, and it also ends January empty. The *optimised* controller is the MILP of Chapter 2 with the import and

export rates of the plan in place of $R_n + N$ and R_n , the same $J = 4$ aging cost, perfect foresight of the net local power, and the same 48-hour rolling horizon, initial energy and terminal constraint as Section ???. It is the like-for-like counterpart of the perfect-foresight spot result: the same optimiser and aging cost on different prices. It may charge from the grid, which is only worthwhile on the time-of-use plan.

Table 7.6: Retail plans, January 2025 (\$). The bill without a battery is \$168.21 on the single-rate plan and \$196.53 on the time-of-use plan.

Plan	Controller	Bill with battery	Gross saving	Rainflow deg.	Net saving	Life lost (%)	Full cycles
Single rate	Self-consumption rule	112.27	55.94	63.36	-7.42	0.528	15.7
Single rate	Optimised, $J = 4$	138.01	30.20	12.38	17.82	0.103	8.5
Time of use	Self-consumption rule	125.32	71.21	63.36	7.85	0.528	15.7
Time of use	Optimised, $J = 4$	134.93	61.61	25.47	36.14	0.212	11.1

The dispatch of the rule is identical on both plans, because it does not depend on price. Grid import falls from 458 kWh to 246 kWh and export from 490 kWh to 252 kWh, with 15.7 equivalent full cycles and 0.53% of cycle life consumed.

The household is better off on the single-rate plan with or without the battery. The choice of plan is worth \$28 in the month without a battery, which is half of the gross saving of the rule on that plan. The battery is nevertheless worth more on the time-of-use plan under either controller, because the imports it displaces are more expensive there. The supply charge is 28% of the single-rate bill and cannot be reduced by storage, so on that plan the battery earns only on the 26.82 c/kWh difference between the usage rate and the feed-in tariff.

The rule has the larger gross saving and the smaller net saving on both plans. It cycles on every solar surplus regardless of the cost of doing so, consumes 0.53% of cycle life in the month, about four times the 0.126% of degradation-aware spot trading, and on the single-rate plan its gross saving does not cover the modelled aging cost. The optimised controller gives up \$10 to \$26 of gross saving, avoids \$38 to \$51 of aging, and returns a positive net saving on both plans. The negative result of the rule is therefore a property of the controller under this aging cost, not of the retail tariff.

Chapter 8

Analysis

8.1 Revenue Analysis

Table 8.1: Value added by the battery over the no-battery household, before aging, by the spot price of the interval in which it was earned (\$).

R_n (\$/MWh)	Intervals	Perfect	Perfect price	Naive	AEMO	LSTM
< 0	1411	2.35	-0.49	0.15	-0.25	-0.07
0 – 50	1211	-1.20	-3.14	-3.04	-3.02	-3.18
50 – 100	3928	2.02	1.40	1.44	1.37	0.79
100 – 300	2338	23.89	20.69	17.22	18.65	19.87
300 – 1000	30	6.57	5.65	5.71	4.24	5.66
> 1000	10	101.53	101.53	101.53	100.56	101.53
Total	8928	135.16	125.64	123.01	121.55	124.60

Three quarters of the value is earned in ten intervals, and every controller captures it. Table 8.1 separates the value added by price band. Of the \$135.16 added under perfect foresight, \$101.53 is earned in the 50 minutes during which the price exceeded \$1000/MWh. All five controllers discharge at the full 11.04 kW in those intervals, with the single exception of the AEMO-driven controller in the \$1052/MWh interval on 14 January, at a cost of \$0.97. This occurs because the price of the current interval is observed before dispatch, and because each controller happened to be holding charge when each spike arrived. None of the forecasters predicted these spikes, and none needed to. The differences between forecasters arise instead in ordinary arbitrage: in the \$100 to \$300/MWh band, where perfect foresight earns \$23.89 against \$17.22 to \$20.69, and at negative prices, where it earns \$2.35 against approximately zero.

8.2 Sensitivity Analysis

8.2.1 Price

8.2.2 Net-Demand

Sensitivity to forecast error

The real forecasters give three points, each with a different kind of error. To measure how dispatch value depends on the *size* of the error alone, the MPC study was repeated with perfect foresight corrupted by synthetic error of controlled magnitude, applied to one input at a time while the other was kept perfect. A forecast issued at interval t for lead $h = 0, \dots, 287$ is

$$\hat{x}_{t+h} = x_{t+h} + \sigma_{\max} \sqrt{\frac{h+1}{288}} e_h, \quad e_h = \rho e_{h-1} + \sqrt{1 - \rho^2} \varepsilon_h, \quad \varepsilon_h \sim \mathcal{N}(0, 1), \quad (8.1)$$

with $e_0 \sim \mathcal{N}(0, 1)$ and $\rho = 0.95$ per five-minute step, so that errors persist for about 1.5 hours rather than alternating between intervals, and grow with lead time to $\sigma_{\max}$ at 24 hours. For price, $x = \sinh^{-1}(R/100)$, the transform used by the LSTM, and the forecast is returned as $100 \sinh(\hat{x})$; the error is therefore approximately additive near zero and proportional during spikes. For net local power, $x = G - A$ in kW. A new error path is drawn at every re-plan, and the current-interval price remains the actual price. Independent errors in each interval were rejected because the MILP would trade on the spurious five-minute spreads they create. One realisation is used per level; each run averages 8928 independent draws.

Table 8.2: Dispatch value under synthetic forecast error, household scenario, $J = 4$, January 2025. Each run perturbs one input; the other is perfect. Value lost is measured against perfect foresight on the same loop.

Input	$\sigma_{\max}$	MAE	Profit at meter (\$)	Rainflow deg. (\$)	Net profit (\$)	Value lost (\$)	EFC
None	–	–	43.31	15.07	28.24	–	9.3
Price	0.25	19.8 \$/MWh	43.68	15.58	28.10	0.14	9.6
	0.5	41.1 \$/MWh	44.67	16.92	27.75	0.49	10.4
	1	95.9 \$/MWh	45.16	20.77	24.38	3.86	12.7
	2	388.3 \$/MWh	34.29	33.80	0.50	27.74	23.1
Net local	0.5 kW	0.27 kW	43.11	15.20	27.92	0.32	9.4
	1 kW	0.53 kW	42.88	15.23	27.64	0.60	9.5
	2 kW	1.06 kW	42.37	15.33	27.04	1.20	9.6
	3 kW	1.60 kW	42.01	15.42	26.58	1.66	9.7

Perfect foresight on the corrected data nets \$28.24, which again matches the 48-hour rolling solve of Table 7.3.

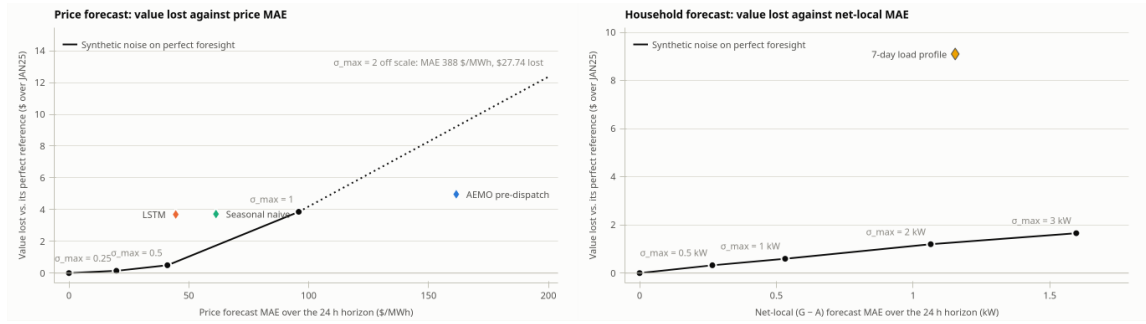


Figure 8.1: Value lost against forecast MAE. The curve is synthetic error on perfect foresight (Table 8.2); the points are the forecasters of Table 7.5, each measured against the reference that isolates its input: the price forecasters against the perfect-price case, the seven-day household profile against perfect foresight.

Unbiased error is cheap until it is large. Price error with an MAE of \$41/MWh, comparable to the LSTM, costs \$0.49, and an MAE of \$96/MWh costs \$3.86. Only at an MAE of \$388/MWh is the value almost eliminated. Household error costs about \$1 per kW of MAE over the range tested.

Price error costs through cycling, not through worse trades. Up to $\sigma_{\max} = 1$ the profit at the meter *rises*, from \$43.31 to \$45.16. The controller acts on spreads that the error has widened, cycling 12.7 equivalent full cycles instead of 9.3, and the additional rainflow degradation of \$5.70 exceeds the additional revenue. The degradation cost in the objective restrains this, but it cannot distinguish a real spread from a forecast one.

The real forecasters lose far more than their MAE predicts. Figure 8.1 places the real forecasters on the synthetic curve. The seven-day household profile has an MAE of 1.15 kW, at which unbiased error would cost about \$1.28; it costs \$9.10. The LSTM and seasonal-naive price forecasts, at \$45 and \$61/MWh, would cost about \$0.71 and \$1.74; each costs about \$3.70. The loss is therefore driven by the structure of the error rather than its size. The synthetic error is unbiased and independent of the data, so it averages out over the month. A real forecast is wrong in a way that depends on the day, for example a seven-day profile on a cloudy day or a price forecast that flattens the daily shape, and such errors move the whole schedule in the same direction. Which features of the error are responsible has not been isolated here. AEMO pre-dispatch sits below the curve, because most of its MAE comes from forecast spikes that did not occur, and these are corrected by the observed current-interval price before they affect dispatch. MAE is therefore a poor guide to the dispatch value of a forecast in either direction, consistent with the loose relation between accuracy and value found above.

8.3 Forecaster Analysis

8.4 Constrained Grid Connection

The result depends on an unconstrained grid connection. No export limit is enforced in the runs above, and during the price spikes the household exports up to 14 kW, the full 11.04 kW of the battery in addition to the solar surplus. Table 8.3 repeats the household $J = 4$ run with the grid export limited to 10 kW and to 5 kW, the latter being a common limit for a single-phase connection. Solar curtailment is not modelled, so under a limit the battery must also keep headroom for any surplus above it. A 5 kW limit removes 40% of the value added by the battery, almost all of it from the spike intervals, and the net profit of the household becomes a net cost. The value outside the spike intervals is nearly unaffected.

Table 8.3: Perfect-foresight MILP, household scenario, $J = 4$, under a grid export limit, January 2025 (\$). Value added is measured against the no-battery household and is before aging.

Export limit	Profit at meter	Rainflow deg.	Net profit	Value added	of which $R_n > 1000$
None	43.31	15.07	28.25	135.11	101.55
10 kW	35.16	15.01	20.14	126.95	93.65
5 kW	-10.63	14.37	-25.00	81.17	49.00

No-battery reference. On spot settlement the household alone pays \$102.04 for imports and receives \$10.24 for exports, a net position of $-\$91.80$ for the month. Household results below are the net position of the whole household, so the value added by the battery is the reported result minus $-\$91.80$.

8.5 Retail self-consumption

8.5.1 Comparison of operating models

Table 8.4 expresses each result as the value added by the battery over the same household with no battery under the same pricing arrangement.

The rows must be compared like for like. The perfect-foresight spot result and the optimised retail results use the same optimiser, the same aging cost and the same foresight, and differ only in the prices: \$120.04 on spot prices against \$36.14 and \$17.82 on the retail plans. The forecast-driven spot results and the self-consumption rule are both realisable, the rule needing no forecast at all: about \$107 on spot prices against \$7.85 and $-\$7.42$.

Table 8.4: Value added by the battery over the no-battery household, January 2025 (\$).

Operating model	Gross value added	Rainflow deg.	Net value added
Spot, perfect foresight, $J = 4$	135.11	15.07	120.04
Spot, LSTM-driven MPC	124.60	17.35	107.25
Spot, seasonal-naive MPC	123.01	15.79	107.23
Spot, perfect foresight, 5 kW export limit	81.17	14.37	66.80
Spot, degradation ignored	186.84	194.62	-7.78
Retail time of use, optimised, perfect foresight	61.61	25.47	36.14
Retail single rate, optimised, perfect foresight	30.20	12.38	17.82
Retail time of use, self-consumption rule	71.21	63.36	7.85
Retail single rate, self-consumption rule	55.94	63.36	-7.42

Comparing a degradation-aware spot controller with the degradation-blind rule would attribute to the tariff a difference in battery wear that belongs to the controller.

On either comparison spot exposure is worth substantially more in this month, and approximately \$100 of the difference is earned in the ten spike intervals. With those intervals excluded the ranking reverses. Perfect foresight on spot prices adds \$33.56 before aging and \$18.49 after it, level with the optimised single-rate result and half of the optimised time-of-use result. The LSTM-driven controller adds \$23.07 before aging and \$5.72 after it, which is within the range of the self-consumption rule. The case for spot exposure in this month therefore rests entirely on the price spikes, and on a grid connection that allows the battery to export at full power during them: under a 5 kW export limit the perfect-foresight spot result falls to \$66.80.

8.5.2 Limitations

The results cover one month, one household and one region. January has the highest solar generation of the year and in 2025 contained one day on which the price reached the market price cap and a second on which it exceeded \$15 000/MWh. The spot results should not be annualised, since three quarters of the value was earned in ten intervals.

The degradation inputs are generic. The replacement cost of \$10 800 is an installed price, which is an upper estimate of the cost of replacing the cells, and the stress function is that of NMC cells, which overstates the aging of a battery built from lithium iron phosphate cells. The sign of the single-rate result of the self-consumption rule and the size of every net figure depend on both. The gross figures do not, although the schedules produced by the degradation-aware MILP do.

Spot exposure is modelled without a retailer. No subscription fee, retail margin or daily supply charge is applied on the spot side, whereas the retail bills include a supply charge of \$46 to \$49. The spot side also omits GST, environmental scheme costs, market fees and

distribution loss factors: the retail rates include GST, while the spot price and the network tariff are used as published. The spot import price is therefore understated by at least 10%, which is visible in the no-battery energy cost of \$91 on spot prices against \$122 on the single-rate plan before the supply charge. The value added by the battery is less sensitive to this than the bills are, since each is measured against its own no-battery case, but every kilowatt-hour the battery imports or displaces is affected. The spot side also uses a flat network tariff, whereas the time-of-use retail plan is built on a time-of-use network tariff.

The price of the current interval is taken as known for the whole interval. In practice it is published at about the start of the interval, and a delay of 30 to 60 seconds in receiving and acting on it would forgo 10% to 20% of the energy discharged in a spike interval.

Apart from Table 8.3, no grid import or export limit is enforced. The household solar is treated as a separate flow that does not share the 11.04 kW inverter of the battery, solar curtailment is not modelled, efficiencies are fixed, and calendar aging is not modelled. The state machine operates a 20 kWh lossless battery, so its result is a qualitative comparison only.

8.6 Observable State Space

Describe the impacts that changing the observable state space for the system demand had and why

Chapter 9

Proposed Methods and Plans

A Gantt chart for part B of this project can be found in Appendix ??.

9.1 Data Analysis

There are two central sources of data considered in this project: 1) load and wholesale price in the NSW NEM; 2) household solar and load data. Ten years of load and wholesale price data have been collated from AEMO. Solar and household load data have been supplied by Ausgrid for 300 homes. The following data study is proposed:

- Examine autocorrelation and residual autocorrelation across network load, price, solar and household load. This will validate our ability to model future scenarios. It is predicted that daily and yearly autocorrelation trends will be present.

9.2 BESS Only Model

This section proposes to replicate existing work on commercial BESS systems to validate the Python framework and data. The system will be modelled as a BESS interacting with the wholesale market with perfect forecasting. The simplified approach has been chosen to develop models and an understanding of the tools available. It will also provide a baseline for future models. The outcomes of this work will be:

- Evaluate and select a model for battery degradation and charge/discharge efficiency.
- Quantify the effect of degradation model fidelity by sweeping the number of cycle depth segments J from the linear case $J = 1$ upwards, comparing dispatch behaviour, revenue and simulated battery life.

- Validate the piecewise linear aging cost against an ex post rainflow cycle count of the resulting SoC profile, to confirm the model reproduces the cycle depths it is intended to approximate and to measure the error as a function of J .
- Refit the cycle depth stress function (3.8) to LFP cycle life data and test the sensitivity of dispatch to the exponent.
- Formulate a basic MDP.
- A reusable Python framework to ingest data, run machine learning models and solve MILP problems.
- A comparison between MILP and ML model performance.

9.3 Household with CER Model

This section of work will model and optimise households operating CER whilst engaging with the NEM wholesale market. It will build upon Section 9.2 by integrating household load and solar data into problem formulation and solutions. It will continue to use the assumption of perfect forecasting and extend this perfect information to household load and solar generation. The outcome of this work will be:

- Formulation of a MDP for households with CER.
- Simulation results for MILP and ML solutions to optimisation.
- A comparison between optimisation techniques to evaluate model performance.

9.4 Price Forecasting

This section of work will develop a model using ML to forecast NEM wholesale pricing, household solar generation and household load. A variety of ML models may be explored including LSTM and XGBoost. This work will then be combined with Section 9.3, to remove the perfect forecasting assumption.

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